

Algebra MATH-310

All information on Moodle

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Organization

Lectures: **Mondays 15:15 - 17:00** SG 1, + live streaming

Exercises: **Mondays 17:15 - 19:00**, SG 1

- + Lectures live streamed and **video recorded**, link on Moodle
- + **Polls** on Ed Forum during class
- + **New: Chat Bot**: AI especially designed for our course - ask any math questions related to the course, see information on Moodle)
- + **Ed Discussion**: questions for the assistants
- + **Old video recordings** online, link on Moodle
- + **Polycopie** written by Joachim Favre, available on Moodle
- + My typed **course notes** on Moodle: **Intro, Integers, Groups, Rings**
- + **Problem sets and solutions** on Moodle

Written exam in January: 100% of the final grade

Idriss Benjelloun
Aleksandra Bogdanova
Raphaël Croué
Mohamed Ennahir
Charbel Raffoul
Mattheo Yang
Svenja Zur Verth (DA)

Plan of the course

- ① Integers: 1 lecture → *today*
- ② Groups: 6 lectures
- ③ Rings and fields: 5 lectures
- ④ Review: 1 lecture

13 lectures

Today: Integers

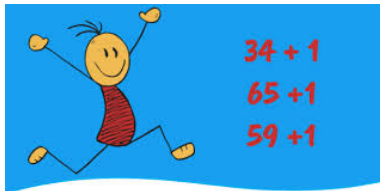
- Ⓐ Induction principle and well-ordering principle
- Ⓑ Prime factorization. Uniqueness.
- Ⓒ Euclidean division. Bézout's theorem
- Ⓓ Euler's totient function

Integers

Question

What is the most basic property of natural numbers?

$$\mathbb{N} = \{0, 1, 2, \dots\}$$



Induction

Question

What is the most basic property of natural numbers?

$$\mathbb{N} = \{0, 1, 2, \dots\}$$

Induction principle

Let $S \subset \mathbb{N}$ such that

- (1) $0 \in S$;
- (2) If $n \in S$, then $n + 1 \in S$.

Then $S = \mathbb{N}$.

Induction vs. Strong Induction

Induction principle

Let $S \subset \mathbb{N}$ such that

- (1) $0 \in S$;
- (2) If $n \in S$, then $n + 1 \in S$. *stronger condition on S*

Then $S = \mathbb{N}$.

weaker
↑
obviously
↑
stronger

Strong Induction principle

Let $S \subset \mathbb{N}$ such that

- (1) $0 \in S$;
- (2) If $\{0, 1, \dots, n\} \subset S$, then $n + 1 \in S$. *weaker condition on S*

Then $S = \mathbb{N}$.

In fact, they are equivalent.

Induction principle

Let $S \subset \mathbb{N}$ such that

- (1) $0 \in S$;
- (2) If $n \in S$, then $n + 1 \in S$.

Then $S = \mathbb{N}$.

Strong Induction principle

Let $S \subset \mathbb{N}$ such that

- (1) $0 \in S$;
- (2) If $\{0, 1, \dots, n\} \subset S$, then $n + 1 \in S$.

Then $S = \mathbb{N}$.

Well ordering principle

Every nonempty subset of $\mathbb{N}$ has a least element.

Poll: Which statement is stronger? **A:** Induction is stronger than Well ordering, **B:** Strong induction is stronger than Well ordering, **C:** Well ordering is stronger than Induction, **D:** All are equivalent

Induction and Well ordering

Proposition ^[PS1] $IP \Rightarrow SIP \Rightarrow WOP \Rightarrow IP$

Induction, Strong induction and Well ordering principles are equivalent.

$IP \Rightarrow SIP$

Let $S \subset \mathbb{N}$: $0 \in S$ and if $\{0, 1, 2, \dots, n\} \subset S \Rightarrow n+1 \in S$

Want to show : $S = \mathbb{N}$ using only IP.

Let $P(n)$ be the statement : $\{0, 1, \dots, n\} \subset S$

Then : $P(0)$ true. : $0 \in S$ given.

If $P(n)$ is true : $\{0, 1, \dots, n\} \subset S \Rightarrow n+1 \in S \Rightarrow \{0, 1, \dots, n+1\} \subset S$
 $\Leftrightarrow P(n+1)$ is true.

$\Rightarrow$ by IP, $P(n)$ is true $\forall n \in \mathbb{N}$

$\Rightarrow \{0, 1, \dots, n\} \subset S \quad \forall n \in \mathbb{N} \Rightarrow S = \mathbb{N}$



Induction and Well ordering

Proposition $WOP \Rightarrow IP$

Induction, Strong induction and Well ordering principles are equivalent.

Suppose $S \subset \mathbb{N}$, $0 \in S$ and if $n \in S \Rightarrow n+1 \in S$. Want to show: $S = \mathbb{N}$

Let $S' = \mathbb{N} \setminus S$. Suppose $S' \neq \emptyset \Rightarrow$ By WOP $\exists k \in S'$ minimal elt ^{using WOP.}
 $k \neq 0$ because $0 \in S \Rightarrow 0 \notin S'$.

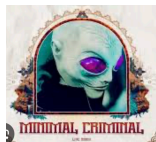
$\Rightarrow k = m+1 \in \mathbb{N}$ for some $m \in \mathbb{N}$, $m \notin S'$ since $k = m+1$ is ^{in S'} minimal
 $\Rightarrow m \in S \Rightarrow$ by condition $m+1 \in S$
on S

$\Rightarrow k = m+1 \in S'$ and $k = m+1 \in S$

contradiction:

$$S \cap S' = \emptyset$$

$\Rightarrow S' = \emptyset \Rightarrow S = \mathbb{N} \quad \square$



proof by "minimal criminal"

Application: prime factorization

Definition

Let $\mathbb{Z} = \{0, \pm 1, \pm 2, \dots\}$. If $a, b \in \mathbb{Z}$ and $a \neq 0$, we say that a divides b if there exists $c \in \mathbb{Z}$ such that $b = ac$.

Definition

A number $p \in \mathbb{Z}_+ = \{1, 2, \dots\}$ is **prime** if $p > 1$ and the only divisors of p are 1 and p . Other integer numbers are composite.

Theorem

Any number $n > 1$ has a prime divisor.

Application: prime factorization

Theorem

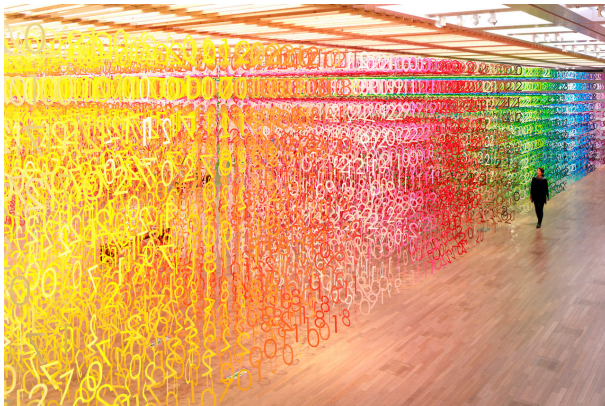
- (1) Any number $n > 1$ is a product of primes; ↖ exercise
- (2) prime factorization is unique. minimal criminal

Let $n = p_1 \dots p_k = q_1 \dots q_m$ the minimal with 2 different prime factorizations; $p_i \neq q_j$
WLOG assume $q_1 > p_1$, let $t = (q_1 - p_1) q_2 \dots q_m > 0$, $t \in \mathbb{N}$, $t < n$
$$t = \underbrace{q_1 q_2 \dots q_m}_{n = p_1 p_2 \dots p_k} - p_1 q_2 \dots q_m = p_1 p_2 \dots p_k - p_1 q_2 \dots q_m = p_1 (p_2 \dots p_k - q_2 \dots q_m) = t$$

 t has a unique prime factorization $\Rightarrow p_1 \mid t$
 $t = (q_1 - p_1) q_2 \dots q_m \Rightarrow p_1 \mid (q_1 - p_1) \Rightarrow q_1 - p_1 = s p_1$, $s \in \mathbb{N}$, $s \geq 1$
 $\Rightarrow q_1 = \underbrace{(s+1)}_{>1} p_1 \Rightarrow q_1$ is not a prime! contradiction
 $\Rightarrow$ no such minimal n can exist
 $\Rightarrow$ every $n > 1$ has a unique prime factorization. ▣

Conclusion: basic properties of $\mathbb{N}$:

- (1) Natural numbers are constructible by induction starting from 0;
- (2) Natural numbers > 1 admit a unique prime factorization.



Euclidean division

Theorem

Let $n \in \mathbb{Z}$ and $d \in \mathbb{Z}_+$. Then there exist two integers $q, r \in \mathbb{Z}$ such that $n = qd + r$ and $0 \leq r < d$. These q, r are unique.

Existence: WOP. Let $S = \{n - kd\}_{k \in \mathbb{Z}} \cap \mathbb{N}$.

$\Rightarrow S \neq \emptyset$, $S \subset \mathbb{N} \Rightarrow$ WOP $\exists r$ the minimal elt of S

$\Rightarrow r = n - kd \Rightarrow n = kd + r$, $r \geq 0$ because $S \subset \mathbb{N}$

If $r \geq d \Rightarrow r' = r - d \in \mathbb{N}$ and $r' = n - kd - d = n - (k+1)d \in S$
 $r' < r$, but r is minimal in $S \Rightarrow$ contradiction.

$\Rightarrow r < d$ and finally $0 \leq r < d$.



Euclidean division

Theorem

Let $n \in \mathbb{Z}$ and $d \in \mathbb{Z}_+$. Then there exist two integers $q, r \in \mathbb{Z}$ such that $n = qd + r$ and $0 \leq r < d$. These q, r are unique.

Uniqueness: Suppose $r_1 + q_1d = r_2 + q_2d$, WLOG $q_1 > q_2$.

$$\underbrace{(q_1 - q_2)}_{\geq 1} d + \underbrace{r_1}_{\geq 0} = \underbrace{r_2}_{\geq 0} \quad \text{contradiction because } r_2 < d.$$

$\underbrace{\hspace{10em}}_{\geq d}$



Definition

If $a, b \in \mathbb{Z}$, then $\text{gcd}(a, b)$ is a positive integer c such that $c|a$ and $c|b$, and if there is another positive integer d with this property, then $d|c$.

Euclidean division

$$n = qd + r$$

Proposition

If $n, q \in \mathbb{Z}$ and $d \in \mathbb{Z}_+$ such that $n = qd + r$, $0 \leq r < d$, then $\gcd(n, d) = \gcd(d, r)$.

c common divisor of $n, d \Rightarrow n = qd + r \Rightarrow c$ divides r

c common divisor of $r, d \Rightarrow n = qd + r \Rightarrow c$ divides n

$\Rightarrow$ The set of common divisors of (n, d)
equals to the set of common divisors of (r, d)

$\Rightarrow \gcd(n, d) = \gcd(r, d)$



proof by staring



Use Euclidean division to find gcd of two numbers

$$d_1 = q_1 d_2 + d_3$$

$$d_2 = q_2 d_3 + d_4$$

...

$$d_{k-1} = q_{k-1} d_k + d_{k+1}$$

$$d_k = q_k d_{k+1} + 0 \quad \Rightarrow \quad d_{k+1} = \gcd(d_1, d_2).$$

Example: $\gcd(432, 315)$

$$\begin{array}{r} 432 \\ 315 \\ \hline 117 \end{array} \quad \begin{array}{r} 315 \\ 1 \\ \hline \end{array}$$

$$\begin{array}{r} 315 \\ 234 \\ \hline 81 \end{array} \quad \begin{array}{r} 117 \\ 2 \\ \hline \end{array}$$

$$\begin{array}{r} 117 \\ 81 \\ \hline 36 \end{array} \quad \begin{array}{r} 81 \\ 1 \\ \hline \end{array}$$

$$\begin{array}{r} 81 \\ 72 \\ \hline 9 \end{array} \quad \begin{array}{r} 36 \\ 2 \\ \hline \end{array}$$

$$\begin{array}{r} 36 \\ 0 \\ \hline 9 \end{array} \quad \begin{array}{r} 9 \\ 4 \\ \hline \end{array}$$

$$\Rightarrow \gcd(432, 315) = 9$$

Applications of Euclidean division

Corollary 1

If $a, b \in \mathbb{Z}_+$, then there exist $x, y \in \mathbb{Z}$ such that $\gcd(a, b) = ax + by$.

Run the Euclidean algorithm backwards!

Example: $\underline{9} = 81 - 2 \cdot 36 = 81 - 2 \cdot (117 - 81) = 3 \cdot 81 - 2 \cdot 117 =$
 $= 3 \cdot (315 - 2 \cdot 117) - 2 \cdot 117 = 3 \cdot 315 - 8 \cdot 117 =$
 $= 3 \cdot 315 - 8(432 - 315) = 11 \cdot \underline{315} - 8 \cdot \underline{432}$

Corollary 2

If $a, b \in \mathbb{Z}_+$ and $d = \gcd(a, b)$, then the equation $ax + by = c \in \mathbb{Z}$ has a solution $x, y \in \mathbb{Z}$ **if and only if** c is a multiple of d .

If $c = kd \Rightarrow d = ax_0 + by_0$ by Corollary 1 $\Rightarrow kd = c = kx_0a + ky_0b$.

c is always of the form kd : if $d|a$ and $d|b \Rightarrow d|(ax+by) \Rightarrow d|c$

Bezout's theorem

Definition

Two integers a, b are **relatively prime** if $\gcd(a, b) = 1$.

Theorem

*Two numbers $a, b \in \mathbb{Z}_+$ are relatively prime **if and only if** there exist $x, y \in \mathbb{Z}$ such that $ax + by = 1$.*

This is a particular case of Corollary 2. $d = 1$

Conclusion: basic properties of $\mathbb{Z}$:

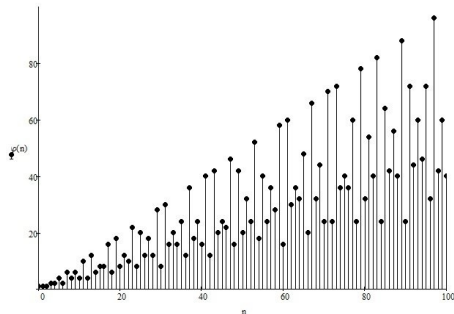
- (1) Euclidean algorithm can be used to find the gcd of two integers
- (2) For two integers a, b we can find two integers x, y such that $xa + yb = c$ if and only if c is a multiple of $\gcd(a, b)$.



Euler's totient function $\varphi(n)$

Definition

For any $n \in \mathbb{Z}_+$ the Euler's totient function $\varphi(n)$ is equal to the number of positive integers k such that $1 \leq k \leq n$ and $\gcd(n, k) = 1$.



$$\varphi(2) = 1$$

$$\varphi(3) = 2$$

$$\varphi(37) = 36$$

$$\varphi(p) = p-1 \quad \text{for any prime } p$$

$$\varphi(15) = 8$$

$$\varphi(p \cdot q) = p \cdot q - q - p + 1 = (p-1)(q-1).$$

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 $\varphi(15)$

Euler's totient function $\varphi(n)$

Let p be a prime.

$$\varphi(p^2) =$$

A: $p^2 - 2p - 1$

B: $p^2 - p$

C: $p^2 - 2p + 1$

D: $p^2 - p - 1$.

$$1 \ 2 \ \dots \ p-1 \ \dots \ p$$

Handwritten sequence of numbers from 1 to p . The numbers 1, 2, 3, ..., $p-1$ are in blue, and p is in red. The sequence is written as $1 \ 2 \ \dots \ p-1 \ \dots \ p$.

$$\varphi(p^2) = p^2 - p \Rightarrow \textcircled{B}$$

$$\varphi(n \cdot m) = \varphi(n) \cdot \varphi(m)$$

if and only if $\gcd(n, m) = 1$.

proof: application of ring theory $\rightarrow$
later in the course

